Making A Large Island (hard)

Problem Statement

Given a 2D binary matrix of size `n x n`, return the integer representing the size of the `largest island` after changing at most `0` to `1`.

Here, `Island` is defined as a `4-directionally` connected group of `1`s.

Examples

- **Example 1**:
  - **Input**:

```
[[1, 0, 1, 0],
 [0, 1, 1, 0],
 [0, 0, 0, 1],
 [1, 1, 0, 0]]
```

  - **Expected Output**: `6`
  - **Justification**: Converting the water cell at position (2,1) will connect two large islands (top right and bottom left) with a new land cell, creating an island of size 6.
- **Example 2**:
  - **Input**:

```
[[1, 1, 1],
 [1, 0, 0],
 [0, 0, 1]]
```

  - **Expected Output**: `6`
  - **Justification**: Changing the cell (1,2) into land will connect the top horizontal island with the bottom right cell, resulting in a single island of size 6.
- **Example 3**:
  - **Input**:

```
[[0, 1, 0],
 [1, 0, 1],
 [0, 1, 0]]
```

  - **Expected Output**: `5`
  - **Justification**: By converting the center cell (1,1) to land, all surrounding cells will connect, forming a cross-shaped island with 5 cells.

Try it yourself

Try solving this question here:

Java

```
import java.util.*;

public class Solution {
    public int largestIsland(int[][] grid) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}

```

Test Case | Run Code Result | Stdout

[[1,0],[0,1]]

Use Example Testcases | Run | Submit

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Solution: Making A Large Island

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